



SELF-SUPERVISED LEARNING FOR PLANT DISEASE CLASSIFICATION: A COMPARATIVE STUDY OF SIMCLR AND BYOL

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This work is not solely the result of my efforts but a tribute to those whose love, strength, and sacrifices have made this journey possible. To my mother, who left her homeland and set aside her dreams so I could pursue mine. Your strength has been the foundation of all I have achieved. To my sister, my constant anchor and guide. Your unwavering belief in me through every challenge has been my greatest source of motivation. To my friends, who lifted me up in times of struggle and celebrated my victories as if they were their own. Your support reminded me that I was never alone. This thesis is dedicated to all of you. With all my heart, thank you. I hope my work honors all that you have given me.

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Abstract

Deep learning has shown strong potential for plant disease detection, but its dependence on large labeled datasets limits practicality due to the high cost and effort required for annotation. Self-supervised learning offers a promising alternative by utilizing unlabeled data, although its application in this field remains limited. This study benchmarks two leading self-supervised methods, SimCLR (contrastive) and BYOL (non-contrastive), using ConvNeXt-Tiny as the backbone and the CCMT dataset. It compares their classification performance and label efficiency against a supervised baseline. The results show that SimCLR achieves a macro F1-score of 0.8675, closely approaching the supervised model's 0.9286, and maintains strong performance even with only 5% of labeled data. BYOL, in contrast, attains an F1-score of 0.7721 and demonstrates less stability when labeled data is scarce. These findings indicate that SimCLR is an effective technique for plant disease detection, while BYOL may require more careful tuning to reach optimal performance. Overall, this work establishes a solid benchmark for future research and highlights the growing value of SSL in plant disease detection.

1 DATA SOURCE, ETHICS, AND TECHNOLOGY STATEMENT

This thesis utilizes the CCMT dataset, which is publicly available through its official repository on [Mendeley Data](#) and authored by Mensah et al. (2023). The dataset is licensed under the Creative Commons Attribution 4.0 International License (CC BY 4.0), which enables reuse and adaptation with appropriate credit. The dataset comprises de-identified plant disease images verified by expert virologists without the involvement of human or animal subjects. While it provides a diverse collection, potential biases may arise from its geographic focus on Ghanaian crops and class imbalance.

The entire codebase was implemented in Python, with a complete list of software libraries and their corresponding versions provided in [Appendix A](#). Portions of the preprocessing pipeline and model architecture were adapted from Lightly's official implementations of SimCLR and BYOL. All modifications were documented and can be found in the following GitHub repository (https://github.com/domdeep/Master_Thesis).

During the development process, GitHub Copilot (<https://github.com/features/copilot>) was used to assist with debugging and refining the code, while Zotero (<https://www.zotero.org/>) was utilized for effective reference management. To enhance the linguistic clarity and coherence of this thesis, ChatGPT-4o (<https://chatgpt.com/>) was used for grammar refinement and stylistic editing, while Grammarly (<https://www.grammarly.com/>) supported additional spelling and grammar correction. Neither tool was used to generate new content or analytical contributions.

The dataset, source code, and third-party figures have been used in full compliance with their respective licenses. All original figures were created by the author unless otherwise stated. External visuals are reproduced under the Creative Commons Attribution 4.0 License, and all third-party materials remain the intellectual property of their original creators.

2 INTRODUCTION

2.1 *Problem Statement and Research Goal*

Fruits and vegetables have long been vital in human nutrition, supplying essential vitamins and proteins across diverse cultures (Slavin & Lloyd, 2012). As the global population grows, food demand increases, making the sustainable production of healthy crops a priority. However, the prevalence of plant disease challenges this goal, significantly threatening agricultural productivity and crop quality (Gai & Wang, 2024). Plant diseases are estimated to cause up to 40% of crop yield losses globally, leading to annual economic damages of approximately \$220 billion (N. Wang et al., 2024). Furthermore, these diseases also raise labor costs and severely impact the livelihoods of smallholder farmers (Vurro et al., 2010).

They arise from various pathogens, including fungi, bacteria, viruses, and parasitic plants (Strange & Scott, 2005). Their emergence and spread are influenced by environmental factors such as temperature, sunlight, and humidity, as well as by agricultural practices and disease management strategies (Gai & Wang, 2024).

Traditionally, early detection of plant infections has relied on farmers visually inspecting crop leaves, which often exhibit texture, color, or structure changes when affected by disease (Al Hiary et al., 2011). While this practice is widespread, it remains subjective, time-consuming, and difficult to scale (Mahlein, 2016). Even trained plant pathologists face challenges in disease detection due to the vast diversity of plant species and the continuous evolution of pathogens (Sladojevic et al., 2016).

In response to these limitations, machine learning and deep learning with convolutional neural networks (CNNs) have emerged as alternatives for plant disease detection (PDD), offering faster and more accurate classification across diverse plant types and infections (Hou et al., 2021; Shoaib et al., 2023).

However, these methods rely heavily on large labeled datasets to learn meaningful features and generalize well. Creating such datasets is costly and time-consuming, often requiring expert annotation, making data availability a significant bottleneck to further improving classification accuracy (Mamun et al., 2024).

Self-supervised learning (SSL) offers a solution to this limitation by enabling models to learn meaningful representations from unlabeled data, thereby reducing the reliance on costly and time-consuming annotation efforts (Ericsson et al., 2022).

While SimCLR (Chen et al., 2020) and BYOL (Grill et al., 2020) have demonstrated state-of-the-art (SOTA) performance on various image clas-

sification tasks, their use in plant disease detection has received limited attention. Research in this area is still emerging, and, to our knowledge, no prior work has directly compared these two self-supervised learning methods within a consistent framework.

2.2 *Scientific Relevance and Societal Relevance*

This thesis addresses this gap by conducting the first comparison between SimCLR (contrastive) and BYOL (non-contrastive) for plant disease detection. Both approaches are trained and assessed using the same dataset, backbone architecture, and training pipeline to ensure fair and reproducible evaluation. The resulting study lays the groundwork for future research in self-supervised learning for PDD.

Although most existing SSL studies in PDD (Bunyang et al., 2023; Chen et al., 2020; Grill et al., 2020) rely on older architecture such as ResNet-50 (He et al., 2015) and VGG16 (Simonyan & Zisserman, 2015), this thesis focuses on three state-of-the-art convolutional neural networks: ConvNeXt-Tiny, EfficientNet-Bo, and DenseNet-121. Evaluating these models provides insights into their overall performance on the CCMT dataset and their effectiveness as backbone networks when combined with SimCLR and BYOL for plant disease detection.

Furthermore, SSL is a potential approach to advancing agricultural analysis, especially in environments where labeled data is scarce. By leveraging vast amounts of unlabeled image and video data, methods such as SimCLR and BYOL enable models to perform complex tasks like field monitoring and robotic harvesting with minimal human annotation. This increases operational efficiency and reduces labor demands and resource waste (Sornapudi & Singh, 2024).

These benefits are especially impactful in precision agriculture, where drone-based systems are widely used for crop health monitoring. While drones can rapidly collect visual data, their full potential is often constrained by the lack of labeled samples (Chin et al., 2023).

2.3 Research Goal

In light of the outlined challenges and the identified research gap, this thesis addresses the following research question:

- **Main RQ:** *To what extent can CNN transfer learning models (e.g., ConvNeXt-Tiny, EfficientNet-Bo, DenseNet-121) serve as effective baseline models for plant disease classification?*
- **SRQ1:** *How does SimCLR improve plant disease classification using the best-performing CNN model from the Main RQ?*
- **SRQ2:** *How does BYOL improve plant disease classification using the best-performing CNN model from the Main RQ?*
- **SRQ3:** *How do SimCLR and BYOL improve label efficiency compared to the best-performing CNN model in Main RQ?*

The main research question investigates how effectively CNNs such as ConvNeXt-Tiny, EfficientNet-Bo, and DenseNet-121 perform as strong baselines for plant disease classification. After identifying the most promising architecture, the study examines whether contrastive self-supervised learning can boost classification performance when applied to this model. It then considers a non-contrastive alternative to determine if similar or better results can be achieved. Finally, the research assesses label efficiency by comparing the performance of both self-supervised approaches and the supervised baseline across different proportions of labeled data.

2.4 Main Findings

Experimental findings indicate that ConvNeXt-Tiny reached the highest downstream output in the fully supervised setting, reaching an F1-score of 0.9286. When pretrained using SimCLR, the model maintained strong performance with an F1-score of 0.8675, notably outperforming the BYOL pretrained counterpart, which attained an F1-score of 0.7721. The t-SNE visualizations further reflected this gap in model output by revealing more distinct class separations for SimCLR. In scenarios with limited training data, SimCLR consistently surpassed BYOL, suggesting a better generalization in PDD. While BYOL extracted some meaningful features, it yielded the weakest results among the evaluated methods and showed the most significant deviation from the supervised ConvNeXt-Tiny baseline. This was further evident in its less coherent clustering patterns observed in the t-SNE visualization.

The ablation study revealed that combining SimCLR pretraining with ImageNet-1k initialization can result in downstream results surpassing the fully supervised ConvNeXt-Tiny model, highlighting the effectiveness of combining self-supervised and transfer learning approaches. BYOL also showed a modest improvement under this setup, though it remained behind SimCLR in model output.

3 LITERATURE REVIEW

3.1 *Traditional Machine Learning for Plant Disease Detection*

Early approaches to plant disease detection primarily relied on classical image processing combined with traditional machine learning techniques. Methods such as color transformations, histogram analysis, and semi-automated segmentation were commonly used to extract handcrafted features from leaf images (Barbedo et al., 2016; Hasan et al., 2022). These features were then used with algorithms such as Support Vector Machines, K-nearest neighbors, and K-means clustering, marking the early integration of machine learning into plant disease detection (Mamun et al., 2024).

For example, Mokhtar et al. (2015) used an SVM to classify two types of viral infections in tomato leaves. In a different approach, Al Hiary et al. (2011) applied K-means clustering to segment diseased areas and subsequently used an artificial neural network for classification based on extracted color and texture features.

3.2 *Deep Learning for Plant Disease Detection*

While conventional machine learning approaches have shown promise in plant disease detection, their reliance on expert knowledge and manual feature extraction often reduces their effectiveness outside controlled settings, limiting their practical use in real agricultural environments (Lu et al., 2021).

In contrast, deep learning has emerged as a more precise and robust alternative. Unlike traditional methods, CNNs automatically learn hierarchical representations directly from raw image data. This ability allows CNNs to capture subtle differences in leaf color, texture, and structure, which are crucial for accurate plant disease identification (Mamun et al., 2024).

Consequently, CNNs have demonstrated strong results across various plant disease classification tasks. For example, EfficientNet-Bo has been used to identify corn leaf diseases with accuracy rates nearing 99%, outper-

forming models such as DenseNet-121 and ResNet-50 (Rajeena P. P. et al., 2023).

Similarly, customized ConvNeXt-Tiny architectures have reached over 97% accuracy in classifying mango leaf infections (Rohman et al., 2024), while DenseNet-121 has proven highly effective in detecting tomato leaf conditions, achieving performance levels above 98% (Vengaiah & Konda, 2023).

3.3 *Self-Supervised Learning*

Deep learning has brought notable advancements to plant disease detection, though its reliance on labeled data continues to pose a challenge. Self-supervised learning addresses this challenge by allowing models to extract meaningful representations from unlabeled data through a two-step process: a pretext task for feature learning and a downstream task for fine-tuning.

During the pretext task, the model learns by solving auxiliary objectives such as estimating missing image patches or reconstructing color information from surrounding pixels. In natural language processing, a similar approach involves predicting omitted words in a sentence using pseudo-labels derived from the input data (Lee et al., 2021).

Although these tasks are not used directly for downstream applications, they guide the model to learn meaningful representations by capturing underlying structures and patterns in the data rather than focusing on task-specific labels (Charoenpanyakul et al., 2024).

After the pretext task, the model has developed a rich internal representation of the data. These learned features can be fine-tuned for downstream tasks like image classification or object detection (Attri et al., 2023).

In general, Self-supervised learning methods are categorized into three main groups: generative, predictive, and contrastive approaches, with non-contrastive methods often viewed as a subcategory within contrastive learning (Mamun et al., 2024).

3.4 *Generative and Predictive Self-Supervised Learning*

Generative self-supervised learning methods focus on reconstructing or creating input data in order to uncover meaningful patterns within it. Notable examples include Generative Adversarial Networks and Masked Autoencoders. In a Generative Adversarial Network, a generator produces synthetic data samples, while a discriminator attempts to differentiate between real and generated examples. This adversarial setup encourages the

generator to produce outputs that closely resemble real data (Goodfellow et al., 2014).

Masked Autoencoders take a different approach. They obscure random parts of an input image and train the model to recover the missing regions. This process helps the model understand the overall visual structure and learn informative features (He et al., 2021).

In comparison, predictive methods involve learning from modified or incomplete data by solving structured tasks. A typical example is the Jigsaw puzzle task, where an image is divided into shuffled patches. The model learns to reassemble the original image, which helps it capture spatial relationships and contextual information (Noroozi & Favaro, 2017).

3.5 *Contrastive Self-Supervised Learning*

Another category of SSL is contrastive self-supervised learning, which aims to learn meaningful representations by comparing distinct data samples. The core idea is to minimize the distance between positive pairs, typically different augmented views of the same image, and maximize the distance between negative pairs, which consist of other images and their augmentations (S.-C. Huang et al., 2023).

3.5.1 *SimCLR*

SimCLR (Simple Framework for Contrastive Learning of Representations), introduced by Chen et al. (2020)¹, is a widely recognized method in contrastive self-supervised learning. It learns useful visual representations by aligning augmented views of the same image (positive pairs) closer in the embedding space while pushing apart representations of different images (negative pairs). Unlike earlier methods such as Wu et al. (2018) and He et al. (2020), which rely on memory banks or momentum encoders to maintain a large pool of negatives, SimCLR eliminates the need for such structures by using large batch sizes (≥ 256) to generate sufficient negative examples during training.

¹ Throughout this thesis, SimCLR refers to the original v1 framework from Chen et al., 2020, not the later v2 version.

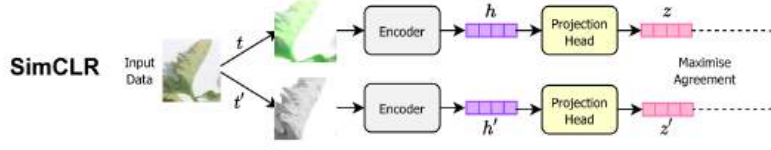


Figure 1: Adapted visualization of the SimCLR framework based on Shah et al. (2021).

The SimCLR framework consists of four main components, as outlined by Chen et al. (2020):

1. **Data Augmentation:** Each image x is augmented twice using two different transformation pipelines, producing two views v and v' . These views form a positive pair, while all other views in the batch are treated as negatives. For example, a batch size of 256 produces 512 augmented views and 256 corresponding positive pairs.
2. **Encoding:** Both views are passed through a shared encoder network $f(\cdot)$ to produce feature representations h_i and h_j .
3. **Projection:** The encoder outputs are then passed through a projection head $g(\cdot)$, typically a two-layer MLP with BatchNorm and ReLU activations, resulting in projected vectors z_i and z_j .
4. **Contrastive Loss:** The NT-Xent (Normalized Temperature-scaled Cross-Entropy) loss is used to bring positive pairs closer and push apart negatives within the embedding space.

The NT-Xent loss is defined as:

$$\ell_{i,j}^{\text{NT-Xent}} = -\log \left(\frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbf{1}_{[k \neq i]} \exp(\text{sim}(z_i, z_k)/\tau)} \right) \quad (1)$$

where:

- $\text{sim}(z_i, z_j)$ is the cosine similarity between representations z_i and z_j ,
- τ is a temperature parameter that controls concentration.

The authors pretrained on ImageNet using a ResNet-50 backbone with a two-layer MLP projection head. Batch Normalization and ReLU were applied after each layer except the last. The training lasted up to 1000 epochs with batch sizes from 256 to 8192, using the LARS optimizer, linear learning rate scaling, a weight decay of 10^{-6} , and a cosine annealing schedule with a 10-epoch warm-up phase.

3.5.2 *Advantages and Challenges of SimCLR*

In their ablation studies, Chen et al. (2020) observed that using deeper and wider backbone networks in SimCLR led to stronger representation learning, with the performance improvements being more pronounced in the self-supervised setting than in the supervised one. Additionally, they found that larger batch sizes significantly enhanced model output during shorter training durations (e.g., 100 epochs), likely due to the increased diversity of negative samples per batch. Although the outcome difference between batch sizes diminished with longer training, smaller batches continued to underperform. Prolonged training also enhanced the model’s output by gradually introducing a wider variety of negative samples.

A central challenge in self-supervised learning is representation collapse, where the network learns a trivial solution by mapping all inputs to identical feature embeddings (Jing et al., 2022). SimCLR effectively addresses this issue through a contrastive loss that simultaneously aligns augmented views of the same image while repelling views of different images, thereby maintaining representational diversity.

Alongside its strengths, SimCLR also has inherent weaknesses such as remaining sensitive to augmentation strategies. For instance, removing color distortion from the augmentation pipeline significantly decreases accuracy, underscoring the importance of carefully designed transformations in contrastive learning. Although this framework avoids the architectural complexity of memory banks, SimCLR introduces significant memory demands due to its reliance on large batch sizes to generate a sufficient number of negative pairs.

3.6 Non-Contrastive Learning

Compared to contrastive learning approaches, non-contrastive methods aim to bring positive pairs closer in the latent space without relying on negative samples (Garrido et al., 2022).

3.6.1 BYOL

BYOL (Bootstrap Your Own Latent) is a non-contrastive self-supervised learning method that employs two neural networks: an online network and a target network. Both share identical architecture and process different modified views of the same image but maintain separate weights. The online network consists of a backbone encoder, a projection head, and a predictor, while the target network includes only the encoder and projection head. During training, the online network is optimized to predict the representation produced by the target network for a differently augmented view. To prevent collapse and stabilize learning without using negative samples, BYOL introduces architectural asymmetry: it adds a predictor only in the online network, applies a stop-gradient to the target network's output, and updates the target network using an exponential moving average (EMA) of the online network's parameters (Grill et al., 2020).

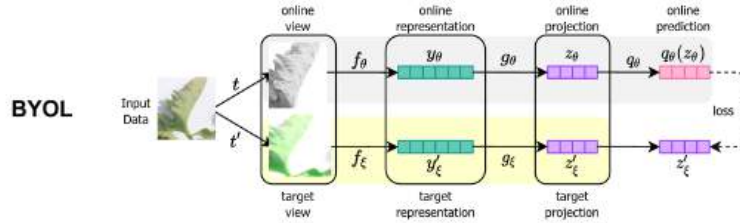


Figure 2: Adapted visualization of the BYOL framework based on Shah et al. (2021).

The BYOL framework consists of six main components, as outlined by Grill et al. (2020) :

1. **Data Augmentation:** Similar to SimCLR, two augmented views of the same image x are generated using stochastic augmentation functions $\mathcal{T}$ and $\mathcal{T}'$, resulting in the online view v and target view v' .
2. **Encoding:** The online encoder f_θ processes the online view v to produce a representation $y_\theta = f_\theta(v)$, while the target encoder f_ξ processes the target view v' to produce $y'_\xi = f_\xi(v')$.

3. **Projection:** Each representation is passed through its respective projection head: the online projection head g_θ maps y_θ to $z_\theta = g_\theta(y_\theta)$, and the target projection head g_ξ maps y'_ξ to $z'_\xi = g_\xi(y'_\xi)$.
4. **Prediction:** The online network further processes z_θ through a predictor q_θ to obtain $q_\theta(z_\theta)$. This predicted vector is compared to the target projection z'_ξ . The target network does not include a predictor, introducing an architectural asymmetry that enables learning without the use of negative samples.
5. **Loss Function:** BYOL minimizes the mean squared error between the ℓ_2 -normalized online prediction and the target projection. The loss for is defined as:

$$\mathcal{L}_{\theta,\xi} \triangleq \left\| \overline{q_\theta(z_\theta)} - \overline{z'_\xi} \right\|_2^2 = 2 - 2 \cdot \frac{\langle q_\theta(z_\theta), z'_\xi \rangle}{\|q_\theta(z_\theta)\|_2 \cdot \|z'_\xi\|_2}$$

6. **Optimization:** Only the online network parameters θ are updated using gradient descent. The target network parameters ξ are updated as an exponential moving average of θ :

$$\xi \leftarrow \tau \cdot \xi + (1 - \tau) \cdot \theta$$

where τ is the momentum coefficient.

BYOL was pretrained on ImageNet using a ResNet-50 backbone for 1,000 epochs with batch sizes up to 4,096, following a protocol similar to SimCLR. It used the LARS optimizer, linear learning rate scaling, cosine decay, and a 10-epoch warm-up. Unlike SimCLR, BYOL omits batch normalization after the projection head and adds a two-layer MLP predictor with batch normalization and ReLU after the hidden layer.

3.6.2 Advantages and Challenges of BYOL

Unlike SimCLR, BYOL learns rich visual representations without relying on negative pairs or large batch sizes, significantly lowering computational requirements. Moreover, BYOL has demonstrated greater robustness to augmentation strategies and scales effectively with deeper architectures and longer training schedules, making it a more flexible and efficient alternative to contrastive SSL methods (Grill et al., 2020).

3.7 Self-Supervised Learning in Plant Disease Detection

Self-supervised learning methods for plant disease detection encompass generative, predictive, contrastive, and non-contrastive approaches. Generative models such as LeafGAN transform healthy leaf images into diseased

ones, improving cucumber disease classification accuracy by 7.4% compared to CycleGAN by focusing on leaf region modifications (Cap et al., 2020). Masked Autoencoders paired with Vision Transformers and weak supervision also show strong performance, achieving 92.9% accuracy in apple leaf disease classification (H. Tian & Song, 2025). On the predictive front, Breiki et al. (2021) applied a Jigsaw task to cassava leaf images, reaching 67% accuracy by training the model to reorder shuffled patches.

Contrastive methods like SimCLR have attracted notable attention in plant disease detection due to their demonstrated effectiveness. For instance, H. Zhang et al. (2021) achieved 90% accuracy on cassava leaf classification using only 70% of labeled data, closely matching the 91.59% of a fully supervised ResNet-18. Likewise, Bunyang et al. (2023) reported varying accuracies for SimCLR across different backbones such as VGG16, ResNet-50, EfficientNet-B0, and MobileNetV3 ranging from 82.48% down to 53.84%. In contrast, non-contrastive methods such as BYOL have been less explored in this field. One exception is Monowar et al. (2022), who benchmarked BYOL using a Siamese deep CNN for clustering fruit leaf diseases, with accuracy varying by crop: 88.7% for apple, 84.5% for corn, and 87.9% for tomato.

Despite growing interest in SSL for plant disease detection, comprehensive comparative evaluations of contrastive and non-contrastive methods remain scarce. Most prior work focuses on individual methods or lacks standardized benchmarking, leaving a significant gap in understanding their relative advantages in this field. The performance variations across studies indicate a need for controlled experiments to evaluate label efficiency and model robustness.

Given SimCLR's stronger empirical record and BYOL's emerging promise despite limited application, this thesis focuses on these two prominent methods. Both share architectural similarities and can be implemented within a unified training framework, enabling a fair and controlled comparison of contrastive and non-contrastive paradigms in plant disease classification where such systematic benchmarking is lacking.

4 METHODOLOGY

This methodology section covers the dataset, preprocessing steps, class distribution, and the CNN architectures used for benchmarking. The self-supervised methods SimCLR and BYOL are introduced later and described in detail in Subsection 5.5.

4.1 *Dataset Description*

This thesis employs the 2023 CCMT Dataset for Crop Pest and Disease Detection, which consists of 24,881 raw and 102,976 augmented color images in JPEG format, distributed across 22 distinct classes. The photos were captured at varying resolutions, ranging from 400×400 to 4032×3024 pixels. The raw dataset includes 7,508 Cassava, 6,549 Cashew, 5,389 Maize, and 5,435 Tomato pictures, totaling approximately 1.22 GB in compressed form. Data collection took place between October and December 2022 on local farms in Ghana. Images were taken from multiple angles (front and back) and under diverse environmental conditions to ensure variability. The annotation was performed manually by expert plant virologists and pathologists, ensuring high quality and reliable labels (Mensah et al., 2023).

4.2 Data Preprocessing



Figure 3: Representative samples from the CCMT dataset after preprocessing: (a) cashew anthracnose, (b) cashew gumosis, (c) cashew healthy, (d) cashew red rust (e) cassava bacterial blight, (f) cassava brown spot, (g) cassava healthy (h) cassava mosaic (i) maize healthy, (j) maize leaf blight, (k) maize leaf spot, (l) maize streak virus (m) tomato healthy, (n) tomato leaf blight, (o) tomato leaf curl, (p) tomato septoria leaf spot, (q) tomato verticillium wilt

Although the dataset was officially reported to contain 24,881 images, exploratory data analysis identified 25,220 files. This discrepancy likely stems from additional files inadvertently included in the dataset directory, possibly due to residual augmented images or duplicate entries not removed during preprocessing. Corrupted files, duplicate entries, and unrelated insect classes (e.g., leaf miner, green mite) were removed to ensure data quality and consistency. In addition, class labels were reformatted by combining the crop type and disease name into a single string (e.g., Tomato_Septoria_leaf_spot), creating unique identifiers that streamlined training, visualization, and metric tracking, all without modifying the original file structure.

4.3 Class Imbalance

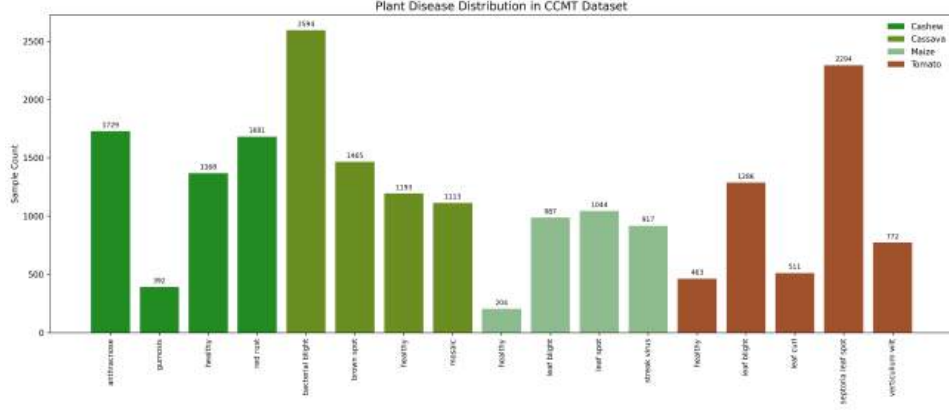


Figure 4: Class distribution in the preprocessed CCMT dataset.

Further analysis revealed a notable class imbalance in the CCMT dataset. As shown in Figure 4, classes such as tomato septoria leaf spot (2,294 images) and cassava bacterial blight (2,594 images) are heavily overrepresented, together comprising over 25% of the dataset. In contrast, underrepresented classes like maize healthy (204 images), cashew gummosis (392 images), and tomato healthy (463 images) account for less than 5%, indicating a pronounced skew in class distribution.

While SSL methods demonstrate increased robustness to class imbalance compared to supervised approaches (H. Liu et al., 2022), the issue is still prevalent in the CCMT dataset. It can result in biased predictions, often favoring dominant categories while misclassifying underrepresented ones (Johnson & Khoshgoftaar, 2019).

4.4 Baseline Models

To address the main research question, three modern CNNs were selected: EfficientNet-Bo, ConvNeXt-Tiny, and DenseNet-121. Chosen for their efficiency and high effectiveness, they reflect a shift away from older models commonly used in earlier SSL work. This subsection outlines their key features and selection rationale.

4.4.1 *EfficientNet*

EfficientNet is a family of CNNs that introduces a structured approach to scaling model depth, width, and input resolution simultaneously using a compound coefficient ϕ (Tan & Le, 2019). This method applies the formulas α^ϕ , β^ϕ , and γ^ϕ to balance these dimensions, enabling efficient model growth without overfitting, vanishing gradients (Bengio et al., 1994), or excessive computational costs. As a result, EfficientNet models registered competitive accuracy while maintaining a lower parameter count and reduced computational cost. This study employs EfficientNet-Bo, the smallest variant in the EfficientNet family, which achieved a top-1 accuracy of 77.1% on ImageNet, surpassing ResNet-50 at 75.3% and DenseNet-169 at 76.2%.

4.4.2 *ConvNeXt*

ConvNeXt, introduced by Z. Liu et al. (2022), is a convolutional architecture designed to modernize traditional CNNs by systematically integrating successful components from Vision Transformers (ViT). Rather than relying on global self-attention, it retains the inductive biases of convolution, such as translation equivariance and local connectivity incorporating ViT modifications like large kernel sizes (e.g., 7×7), GELU activation, Layer Normalization, and inverted bottlenecks. These architectural updates allow ConvNeXt to close the efficacy gap between pure CNNs and transformer models while maintaining computational efficiency on image classification tasks. In particular, the ConvNeXt-Tiny variant demonstrates that, with proper architectural tuning, it can attain 82.1% top-1 accuracy on ImageNet-1K, surpassing ResNet-50 with 75.3%.

4.4.3 *DenseNet*

DenseNet addresses challenges such as vanishing gradients and inefficient feature reuse by employing dense connectivity. In contrast to traditional CNNs, where layers are connected sequentially, DenseNet connects each layer to all preceding layers within a block. This design improves gradient flow, promotes feature reuse, and enhances parameter efficiency. DenseNet-121, a widely adopted variant consisting of 121 layers, achieved a top-1 error rate of 25.02% and a top-5 error of 7.71% on ImageNet-1K, all while maintaining a relatively compact size compared to models like ResNet (G. Huang et al., 2018).

5 EXPERIMENTAL SETUP

This section outlines the experimental setup, including the dataset, model architecture, training procedures, hyperparameter tuning, evaluation metrics, and computational setup. A schematic diagram illustrates the whole workflow, from data preparation to model evaluation.

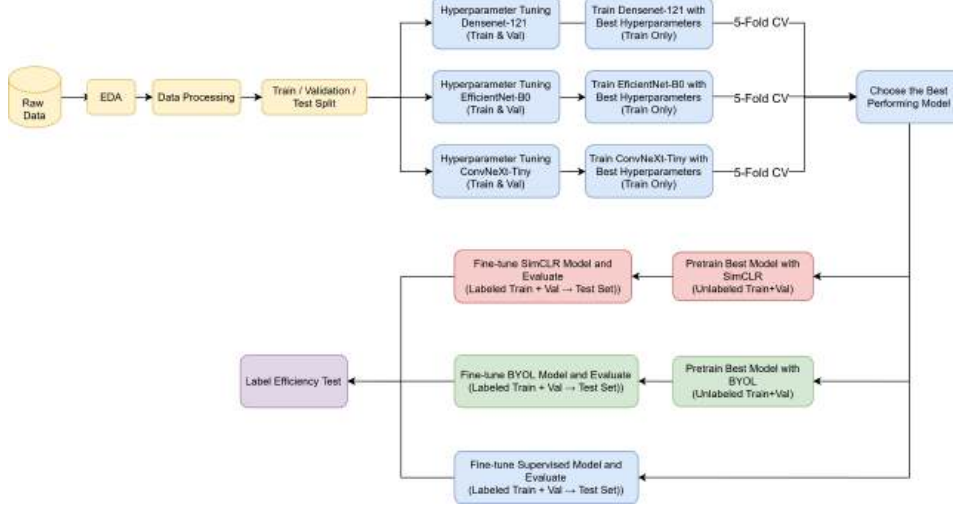


Figure 5: Experimental workflow. Yellow represents data processing and exploratory data analysis (EDA). Blue corresponds to the first step, addressing the main research question through supervised model benchmarking. Red highlights the workflow for the first sub-question involving SimCLR. Green represents the second sub-question focusing on BYOL. The final step, shown in purple, evaluates label efficiency and addresses the third sub-question.

5.1 Dataset Splits and Augmentation

5.1.1 Train/Validation/Test Split

The dataset was divided into training (80%), validation (10%), and test (10%) sets using stratified sampling to ensure class balance across splits. The validation set serves for hyperparameter optimization, while the test set remains strictly held out for final evaluation (Shah et al., 2021). During self-supervised pretraining using SimCLR and BYOL, the training and validation sets (comprising 90% of the data) are merged and utilized without labels. In the subsequent fine-tuning phase, this combined subset is used for supervised learning with labels, and the final models are evaluated on the test set (Su et al., 2021).

5.1.2 Data Augmentation

During the self-supervised pretraining phase, this thesis adopted the data augmentation strategies proposed in the SimCLR and BYOL frameworks.

The SimCLR pipeline served as the foundation and included the following transformations:

- **RandomResizedCrop (224×224)**: Randomly crops and resizes the image to introduce spatial variation.
- **RandomHorizontalFlip (p = 0.5)**: Horizontally flips the image with a 50% probability to introduce orientation diversity.
- **ColorJitter (brightness = 0.8, contrast = 0.8, saturation = 0.8, hue = 0.2)**: Randomly adjusts color properties to enhance invariance to lighting and color changes.
- **RandomGrayscale (p = 0.2)**: Converts the image to grayscale with a 20% probability to further diversify color channels.
- **GaussianBlur (kernel size = 23, $\sigma \in [0.1, 2.0]$)**: Applies a Gaussian blur to reduce local details and emphasize broader patterns.

BYOL adopts a nearly identical augmentation pipeline but uses slightly weaker transformations: brightness, contrast, and hue adjustments are halved, saturation is reduced to 25%, and Gaussian blur is applied to the first view with only a 10% probability. In addition, solarization is applied exclusively to the second view.

For supervised and SSL fine-tuning in this thesis, the augmentation strategy was simplified to random cropping and horizontal flipping to improve stability and reduce overfitting. This minimal setup aligns with SimCLR’s fine-tuning protocol, which also used the same augmentations and reported no significant output gains from applying stronger transformations during this phase.

All training images were resized to 224×224 pixels to match the input size expected by the ImageNet-1K-pretrained CNN models used in the present research. Images were first resized for validation and test sets, so the shorter side was 256 pixels, followed by a center crop to 224×224. This preprocessing strategy is consistent with the original protocols used in the SimCLR and BYOL frameworks.

5.1.3 Data Normalization

All images were normalized using the standard ImageNet-1K mean $[0.485, 0.456, 0.406]$ and standard deviation $[0.229, 0.224, 0.225]$ to match the input distribution expected by models pretrained on ImageNet-1K, helping to reduce distribution skew and enable more efficient learning (Corley et al., 2023).

5.2 Model Architecture and Configuration

5.2.1 Classification Head

The original classification heads of the CNN models were replaced with a standardized and modular fully connected head designed to match the number of target classes. This modification ensured fair and consistent comparisons across different architectures. Each head consists of a flattening layer followed by either a single fully connected (FC) layer or multiple stacked FC layers, with the architecture determined through hyperparameter optimization. In configurations with more than one layer, batch normalization, ReLU activation, and dropout are applied between layers to improve training stability and generalization. Specifically, batch normalization reduces internal covariate shift and accelerates convergence (Ioffe & Szegedy, 2015), while dropout improves generalization by randomly deactivating neurons during training (Srivastava et al., 2014).

5.3 Class Weights and Stratified Sampling

Class weighting and stratified sampling were employed as part of this research to address the issue of class imbalance. Class weights, computed as the inverse of class frequencies, were applied in the loss function to counteract the dominance of frequent classes and highlight underrepresented ones (Cui et al., 2019). In parallel, stratified sampling was applied during the creation of the training, validation, and test splits to preserve the original class distribution. The resulting indices were reused throughout the study, including in K-fold cross-validation, to ensure consistent and accurate model estimates. This approach reduces variance and bias often introduced by standard cross-validation in imbalanced datasets (Kohavi, 1995).

5.4 *Supervised Training*

This section describes the training and optimization of the CNNs using labeled data, including hyperparameter tuning and model assessment. All models were initialized with ImageNet-1K pretrained weights, leveraging transfer learning to provide strong initial feature representations. When fine-tuned, these representations lead to improved performance compared to training the models from scratch (Huh et al., 2016).

The models were trained using cross-entropy loss, and each CNN, equipped with a custom classification head, was fine-tuned end to end with all layers unfrozen. To balance computational efficiency, a batch size of 32 was used for all supervised training and fine-tuning experiments. A larger batch size of 256 was chosen for self-supervised pretraining, as it is the smallest commonly reported in the original self-supervised learning papers. This decision also accounted for hardware limitations while staying aligned with established practices.

5.4.1 *Hyperparameter Tuning*

The first stage of the pipeline involved hyperparameter optimization, conducted independently for DenseNet-121, ConvNeXt-Tiny, and EfficientNet-Bo using the training and validation sets.

Hyperparameter optimization was conducted using the Optuna framework (Akiba et al., 2019), which applies the Tree-structured Parzen Estimator algorithm (Bergstra et al., 2011) to efficiently explore promising regions of the search space. Its flexible design and adaptive sampling strategy make Optuna particularly effective for tuning deep learning models within constrained computational budgets.

For each architecture, 15 Optuna trials were conducted, with training capped at 20 epochs per trial. Early stopping was applied to halt training after three consecutive epochs without improvement in the validation F1-score, reducing overfitting and improving efficiency. Optuna’s median pruning strategy was used as well to terminate underperforming trials early to conserve resources further.

Model effectiveness during tuning was evaluated on the validation set using the macro-averaged F1-score, which gives equal weight to each class and is well-suited for imbalanced classification tasks (Hinojosa Lee et al., 2024). Final hyperparameter configurations were selected based on the highest F1-score obtained during validation.

5.4.2 Hyperparameter Search Space

The hyperparameter search space included the number of hidden units in the classification head, which was treated as a tunable parameter with values of 256, 512, and 1024. Dropout rates between 0.1 and 0.6 were also explored to promote generalization and mitigate overfitting (Srivastava et al., 2014).

In addition, the optimizers Adam (Kingma & Ba, 2017), AdamW (Loshchilov & Hutter, 2019), and RMSprop (Hinton et al., 2012) were selected, as each offers a distinct approach to learning rate adaptation and regularization. The learning rate and weight decay were sampled from log-uniform distributions in the ranges of 1×10^{-5} to 5×10^{-4} and 1×10^{-6} to 1×10^{-3} , respectively.

Table 1: Hyperparameter search space for the three models during optimization.

Model	Hyperparameter search ranges
DenseNet-121, ConvNeXt-Tiny, EfficientNet-Bo	Learning rate: log-uniform $[1 \times 10^{-5}, 5 \times 10^{-4}]$
	Fully connected layers: [1, 2, 3]
	Dropout: [0.1, 0.6]
	Weight decay: log-uniform $[1 \times 10^{-6}, 1 \times 10^{-3}]$
	Hidden dimension: [256, 512, 1024]
	Optimizer: [Adam, AdamW, RMSprop]

Note. No hyperparameter tuning was performed during self-supervised pretraining. For default configuration details, see Section 5.5.

5.4.3 Baseline Evaluation and Model Selection

After hyperparameter tuning, each model was trained using its best-performing hyperparameters² and evaluated employing 5-fold cross-validation on the training set. This method reduces output variability and enhances generalization by rotating the validation split across folds (Raschka, 2018). The training was conducted for up to 25 epochs for every fold, with the test set remaining entirely isolated. The architecture with the highest marco F1-score was chosen as the backbone for SimCLR and BYOL pretraining.

² See Appendix B for a detailed summary of the best hyperparameter configurations from each model.

5.5 *Self-Supervised Pretraining*

SimCLR and BYOL pretraining follow the configurations proposed by Chen et al. (2020) and Grill et al. (2020), with minor modifications to accommodate dataset characteristics and hardware constraints, ensuring consistency and fair comparison.

The ResNet-50 backbone used in the original studies was replaced with ConvNeXt-Tiny, which was identified as the most effective model in the previous CNN benchmark evaluation. Corresponding adjustments were made to the projection head dimensions to match the output size of the new backbone while preserving the original architectural structure and loss function used in the reference implementations.

Both methods were trained for 100 epochs using a batch size of 25. To support stable representation learning, a cosine annealing learning rate schedule was applied, with a linear warm-up during the first 10 epochs. Whereas the original publications used LARS, this study employed Adam due to its proven stability with smaller batches and widespread adoption in self-supervised learning (Kingma & Ba, 2017; X. Wang & Aitchison, 2024). In the context of plant disease detection, Adam has demonstrated strong performance with SimCLR (Bunyang et al., 2023; H. Zhang et al., 2021) and has also been successfully applied in related vision tasks involving BYOL (Ciga et al., 2021; Shi et al., 2020). Therefore, Adam was consistently used across both methods throughout the study to ensure a fair benchmark.

Hyperparameters such as learning rate and weight decay for SimCLR and BYOL were adopted from Ciga et al. (2021), who benchmarked both methods using the Adam optimizer as well. Key settings from the original implementations were preserved, including a temperature of 0.1 for SimCLR and an EMA momentum value of 0.996 for BYOL. No additional hyperparameter tuning was performed.

Although these fixed settings may constrain optimal results, they establish a controlled environment for fair comparison and reproducibility.

5.6 *Fine-Tuning*

Recent work by He et al. (2021) advocates for full fine-tuning as a more reliable evaluation strategy for self-supervised learning. This approach has also been adopted by Zhou et al. (2022) and Dong et al. (2022), who argue that linear probing fails to reflect a model's actual performance on downstream tasks accurately and often underestimates its ability to learn complex, non-linear representations.

Following this approach, all models in this thesis were fine-tuned accordingly. During the fine-tuning stage, the supervised ConvNeXt-Tiny

model, along with the SimCLR- and BYOL-pretrained variants, were initialized with the best hyperparameters identified during the benchmarking stage. Each model was then retrained for 100 epochs using the combined labeled training and validation sets. The final evaluation was performed on a separate test set.

5.7 Algorithms and Software

The supervised experiments were conducted locally on an NVIDIA RTX 3080 Ti, while SSL pretraining was conducted on the GPU4EDU servers at Tilburg University. The self-supervised implementations were adapted from the Lightly library to simplify development while following the original frameworks closely. Code was developed using PyTorch and PyTorch Lightning, with hyperparameter optimization handled via Optuna. Development occurred in Visual Studio Code, and Jupyter Lab was used for visual analysis. On average, pretraining required 60 hours, hyperparameter tuning 15 hours, and fine-tuning 9 hours.³

5.8 Evaluation Metrics

The primary evaluation metric in this thesis is the macro F1-score, as it balances precision and recall and gives equal importance to all classes, making it particularly suitable for imbalanced datasets (Hinojosa Lee et al., 2024; Sokolova & Lapalme, 2009). Macro averaged precision and recall were also reported. At the same time, accuracy was included for reference despite its bias towards the majority classes. Confusion matrices were used to analyze prediction errors and assess consistency between disease categories.

In addition, the learned feature representations were qualitatively analyzed using t-distributed Stochastic Neighbor Embedding (t-SNE). This nonlinear dimensionality reduction technique projects high-dimensional data into a lower-dimensional space while preserving local and global structures. By mapping similar data points nearby and separating dissimilar ones, t-SNE reveals clustering and class separability, with color-coded plots to support visual interpretation (van der Maaten & Hinton, 2008). For this analysis, t-SNE was applied with two output dimensions and a perplexity of 30, which serves as a smooth estimate of the effective number of neighbors. Every other hyperparameter was kept at its default values.

³ For a complete list of libraries, versions, and additional tools used in the project, see Appendix Table 7.

5.9 Label Efficiency

Following the evaluation protocol of Bunyang et al. (2023), the three ConvNeXt-Tiny variants (Supervised, SimCLR, and BYOL) were trained using 5%, 10%, 20%, 40%, 60%, 80%, and 100% of the combined training and validation data. Each experiment was repeated three times with different random seeds, and the macro-average F1-Score was recorded to assess how model effectiveness scales with increasing amounts of labeled data.

6 RESULTS

This section first compares the classification outcomes of the three CNN models. It then examines whether pretraining with SimCLR and BYOL enhances predictive capability through fine-tuning, and evaluates the models' label efficiency. Finally, t-SNE visualizations are presented to support the interpretation of the learned feature representations.

6.1 Results for Main Research Question

Table 2: Supervised model benchmark results using 5-fold cross-validation (mean $\pm$ standard deviation)

Model	F1	Precision	Recall
DenseNet-121	0.8017 $\pm$ 0.0107	0.8187 $\pm$ 0.0099	0.8202 $\pm$ 0.0104
EfficientNet-Bo	0.8735 $\pm$ 0.0128	0.8838 $\pm$ 0.0121	0.8844 $\pm$ 0.0118
ConvNext-Tiny	0.8740 $\pm$ 0.0108	0.8848 $\pm$ 0.0111	0.8851 $\pm$ 0.0103

Table 2 presents the 5-fold cross-validation findings for the three CNN models: DenseNet-121, EfficientNet-Bo, and ConvNeXt-Tiny. ConvNeXt-Tiny reached the highest mean F1-score of 0.8740 ± 0.0108 , marginally outperforming EfficientNet-Bo (0.8735 ± 0.0128). It also attained the best mean precision of 0.8848 ± 0.0111 and recall of 0.8851 ± 0.0103 , indicating strong predictive capabilities. In contrast, DenseNet-121 performed worse, with a mean F1-score of 0.8017 ± 0.0107 , underscoring the advantages of more recent architectures such as EfficientNet and ConvNeXt for this task. Based on these findings, ConvNeXt-Tiny was chosen for further evaluation due to its robust overall capability and slightly lower standard deviations across all metrics. Nonetheless, the narrow discrepancy in model output between ConvNeXt-Tiny and EfficientNet-Bo suggests that both models are

viable options for robust plant disease classification under class imbalance.
4

Table 3: Final test set performance of ConvNeXt-Tiny (Supervised)

Model	F1-score	Precision	Recall	Accuracy
ConvNeXt-Tiny (Supervised)	0.9286	0.9331	0.9253	0.9415

The ConvNeXt-Tiny (Supervised) model attained a strong macro F1-score of 0.9286, with a precision of 0.9331 and a recall of 0.9253, indicating balanced model effectiveness across all 17 plant disease categories. The confusion matrix in Figure 6 shows consistently accurate predictions, with classes such as cashew gummosis, cashew red rust, and cassava mosaic classified with high precision. Misclassifications were limited and primarily occurred among tomato diseases, including leaf curl, leaf blight, and septoria leaf spot. In contrast, confusion among other pairs, such as maize healthy and leaf blight, was minimal.

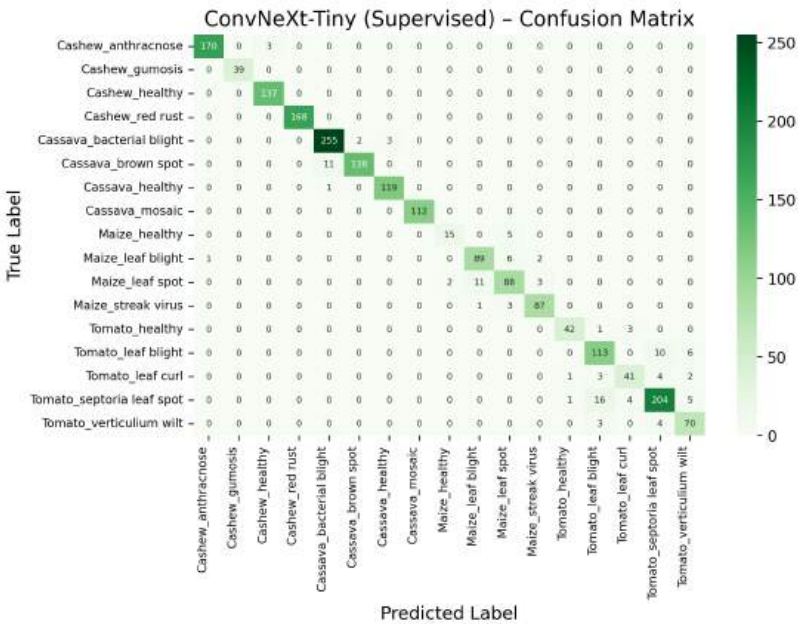


Figure 6: Confusion matrix for ConvNext-Tiny (Supervised)

4 The accuracy metric was not computed in this stage because of computational constraints but will be included in subsequent evaluations.

6.2 Results for Research Question 1

Table 4: Final test set performance of ConvNeXt-Tiny (SimCLR)

Model	F1-score	Precision	Recall	Accuracy
ConvNeXt-Tiny (SimCLR)	0.8675	0.8576	0.8818	0.8786

The ConvNeXt-Tiny model trained with SimCLR recorded a macro F1-Score of 0.8675, with a recall of 0.8818 and precision of 0.8576. As shown in Figure 7, most categories were predicted with high precision. However, tomato diseases proved to be the most challenging. The most pronounced classification errors occurred between tomato leaf curl and tomato septoria leaf spot, with additional mislabeling involving tomato verticillium wilt. In comparison, maize leaf blight and leaf spot also exhibited some ambiguity, though with less frequent misclassification.

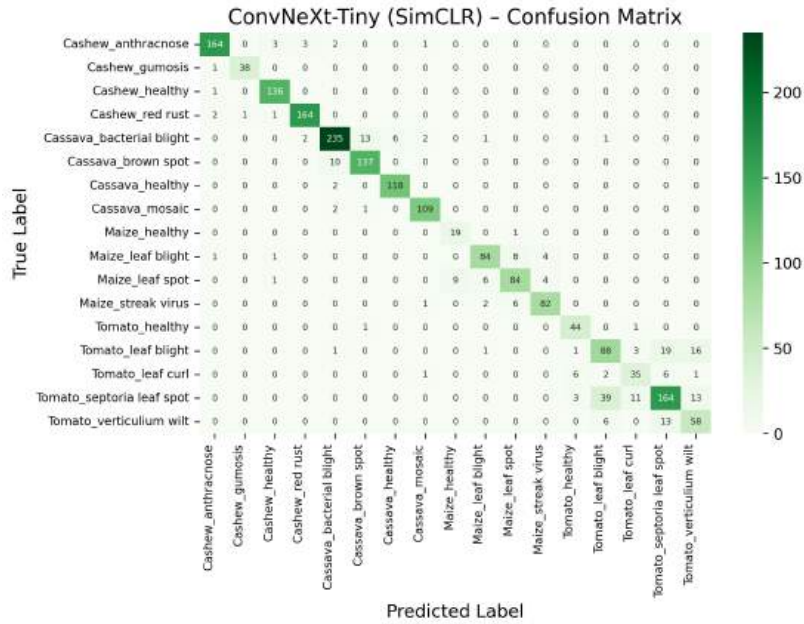


Figure 7: Confusion matrix for ConvNeXt-Tiny (SimCLR)

6.3 Results for Research Question 2

Table 5: Final test set performance of ConvNeXt-Tiny (BYOL)

Model	F1-score	Precision	Recall	Accuracy
ConvNeXt-Tiny (BYOL)	0.7721	0.7657	0.723	0.8051

ConvNeXt-Tiny (BYOL) achieved a macro F1-Score of 0.7721, with a precision of 0.7657 and a recall of 0.723, reflecting a decline in overall performance compared to both SimCLR and the supervised baseline. Despite the absence of labels during pretraining, the model still exhibited competitive generalization after fine-tuning.

As illustrated in Figure 8, the confusion matrix indicates generally accurate predictions but with more prominent classification inconsistencies among the tomato disease categories relative to the SimCLR model. Significant overlaps were observed between tomato leaf curl, septoria leaf spot, and verticillium wilt. Additionally, moderate misidentifications between maize leaf blight and leaf spot remained consistent with earlier patterns.

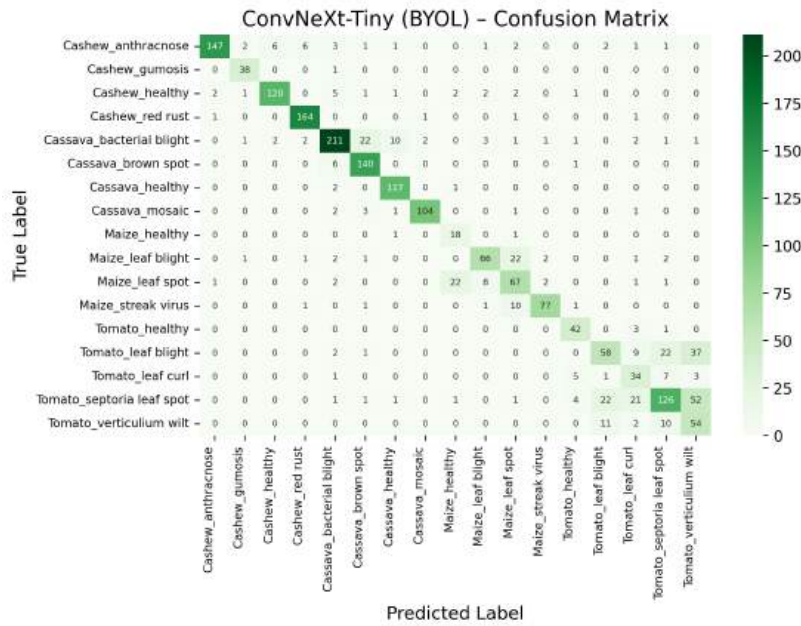


Figure 8: Confusion matrix for ConvNext-Tiny (BYOL)

6.4 Comparison of Supervised and Self-Supervised Models

Table 6: Comparison of test set results for ConvNeXt Tiny models

Model	F1-score	Precision	Recall	Accuracy
ConvNeXt-Tiny (Supervised)	0.9286	0.9331	0.9253	0.9415
ConvNeXt-Tiny (SimCLR)	0.8675	0.8576	0.8818	0.8786
ConvNeXt-Tiny (BYOL)	0.7721	0.7657	0.7230	0.8051

Across all evaluated models, the supervised ConvNeXt-Tiny produced the highest outcomes across all metrics, with a macro F1-Score of 0.9286, precision of 0.9331, and recall of 0.9253. These findings highlight the effectiveness of supervised learning in capturing fine-grained distinctions between plant disease classes.

The SimCLR pretrained model followed closely, reaching a macro F1-Score of 0.8675. Despite being trained without labels during pretraining, SimCLR demonstrated strong generalization after fine-tuning. Most disease categories were predicted with high precision. However, classification errors were more frequent when contrasted with the supervised baseline, especially within the tomato classes, where overlaps between leaf curl, septoria leaf spot, and verticillium wilt were observed. Minor misidentifications between maize leaf blight and leaf spot also persisted.

The BYOL pretrained model registers the lowest evaluation outcome, with a macro F1-Score of 0.7721, precision of 0.7657, and recall of 0.7230. While it still demonstrated reasonable generalization, the model had increased difficulty distinguishing between the tomato disease categories. Pronounced classification inconsistencies and overlapping visual symptoms limited its ability to separate certain conditions, resulting in lower F1-scores than SimCLR and the supervised version.

6.5 Dimensionality Reduction (*t*-SNE)

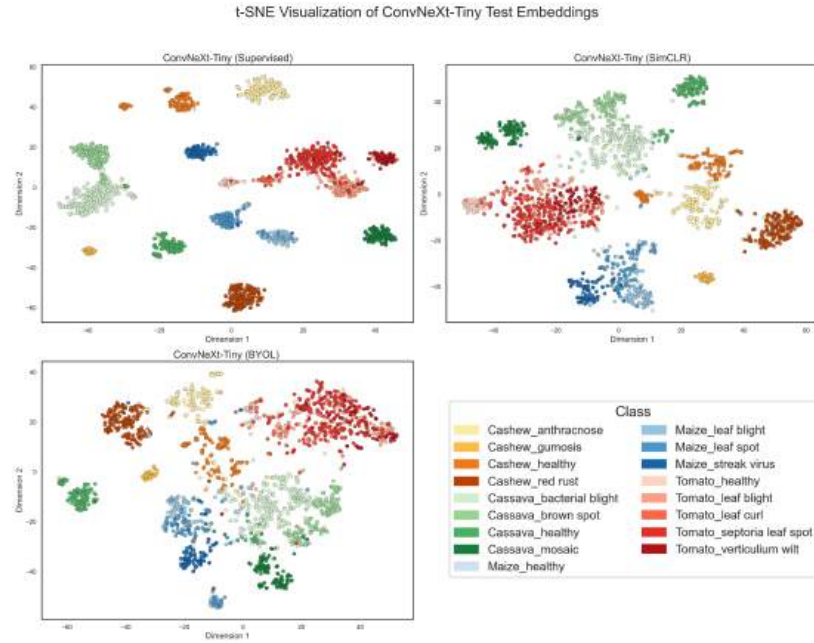


Figure 9: Representative t-SNE visualizations for each model.

The t-SNE plots in Figure 9 reveal apparent differences in how each ConvNeXt-Tiny model clusters its learned feature representation.

The supervised model creates the most distinct clusters, with many categories forming compact and separated groups. Some overlap remains, especially among visually similar types such as tomato leaf curl and tomato septoria leaf spot, underscoring the challenge of making precise distinctions even under full supervision.

In comparison, the SimCLR model produces more scattered clusters with noticeable blending, suggesting that although it captures the overall structure, its features are less aligned with class boundaries. BYOL, on the other hand, exhibits the least structured space, with significant class mixing and weak group formation, especially among cassava and tomato diseases.

In summary, the supervised model yields the strongest structural separation, followed by SimCLR, which demonstrates partial alignment with class boundaries. BYOL, in contrast, demonstrates limited class separation. These observations are consistent with the earlier findings in the results section, reinforcing the connection between feature structure and classification ability.

6.6 Results for Research Question 3

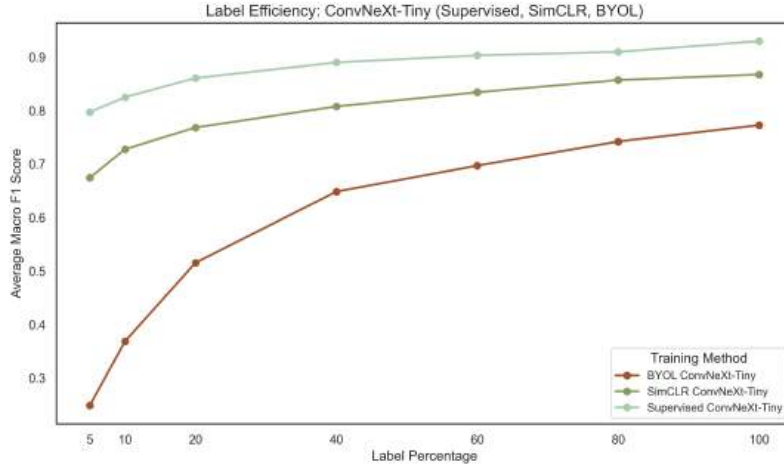


Figure 10: Label efficiency comparison across all models

Figure 10 compares the label efficiency of three ConvNeXt-Tiny variants: Supervised, SimCLR, and BYOL. The Supervised model consistently reached the highest macro F1-scores across all label fractions, reaching 0.80 with only 5% of labeled data and improving to around 0.90 at 80%. SimCLR also demonstrates robust results, particularly in low-label scenarios. It starts at approximately 0.68 with 5% of the training data and increases to around 0.80 as more annotations are introduced. Although SimCLR lags behind the fully supervised model, it shows robust generalization and effectively leverages limited labeled data.

In contrast, BYOL shows noticeably lower capability, particularly under limited supervision. Starting at 0.25 with 5% of labeled data, it gradually increases to 0.70 at 80%, consistently underperforming compared to both SimCLR and the supervised baseline. This evaluation discrepancy suggests that BYOL may struggle to extract useful representations in low-label settings without additional tuning or access to more labeled data.

These findings highlight that while supervised learning yields the strongest label efficiency, SimCLR presents a competitive alternative under limited data, whereas BYOL’s effectiveness remains more constrained.

7 DISCUSSION

Motivated by the high cost of manual annotation, this thesis investigated whether self-supervised learning can achieve results comparable to supervised models in plant disease classification. SimCLR and BYOL performed close to the supervised baseline, with SimCLR generally showing stronger results. These findings emphasize the promise of self-supervised methods as practical alternatives in settings with scarce annotations.

7.1 *Main Research Question: Evaluating CNN Baselines for Plant Disease Classification*

The conclusions of this study show that modern CNN architectures can serve as effective baselines for plant disease classification. Among the three evaluated models, ConvNeXt-Tiny, EfficientNet-Bo, and DenseNet-121, ConvNeXt-Tiny obtained the highest macro F1-score of 0.9286, slightly outperforming EfficientNet-Bo. DenseNet-121 showed weaker classification quality than the other models but achieved reasonable performance metrics. The better outcomes of ConvNeXt-Tiny and EfficientNet-Bo may be attributed to their recency and architectural design, which allow for more effective representation learning in challenging tasks such as plant disease detection.

7.2 *Sub-Question 1: Impact of SimCLR on Plant Disease Classification*

Previous research has shown that SimCLR can come close to the performance of supervised models in plant disease classification. Bunyang et al. (2023) tested SimCLR with models like ResNet-50 and VGG16 on the PlantVillage dataset, achieving 81.90% and 82.48% accuracy, slightly below their supervised baselines of 82.35% and 84.00%. Their experiments were based on accuracy, used older architectures, and did not involve hyperparameter tuning. H. Zhang et al. (2021) also demonstrated promising results, reaching 90.0% accuracy on cassava leaf classification using only 50% labeled and 20% unlabeled data, nearly matching their supervised baseline of 91.59% trained on 70% labeled data. In this thesis, SimCLR is applied to the CCMT dataset using ConvNeXt-Tiny and evaluated with macro F1-score. Fine-tuning with all labeled data resulted in an F1-score of 0.8675, compared to the supervised score of 0.9286. While differences in datasets and evaluation metrics limit direct comparison, the results confirm a clear trend: SimCLR narrows the gap to supervised learning and may benefit from more recent architectures and carefully tuned pipelines.

However, the ablation study (see [Appendix C](#)) demonstrated that combining SimCLR pretraining with ImageNet-1k weights can surpass the fully supervised model.

7.3 *Sub-Question 2: Impact of BYOL on Plant Disease Classification*

Among all models evaluated, BYOL with the ConvNeXt-Tiny architecture produced the weakest results, reaching a macro F1-score of 0.7721. This was notably lower than the supervised ConvNeXt-Tiny model at 0.9286 and the SimCLR-pretrained version at 0.8675.

BYOL showed limited generalization and struggled to distinguish between tomato and maize diseases. These issues were evident in the confusion matrix and t-SNE plots, which revealed more overlapping class clusters than SimCLR and the supervised baseline. However, the ablation study indicates that pretraining on a larger dataset like ImageNet-1k could improve BYOL’s performance.

Although BYOL has surpassed SimCLR on widely used datasets like ImageNet-1K and CIFAR-10, these outcomes were obtained under different experimental setups, often involving deeper models such as ResNet-50, larger batch sizes, and optimization algorithms like LARS (Grill et al., 2020; Marks et al., 2024). In contrast, the current work used a more compact setup that included ConvNeXt-Tiny, the Adam optimizer, a batch size of 256, and a shorter training schedule. The observation that BYOL underperformed compared to SimCLR in this context aligns with the findings of Ciga et al. (2021), who also reported stronger results for SimCLR in similar application domains. Like the present study, their work suggests that performance gains observed on large-scale benchmarks may not necessarily carry over to smaller datasets, such as the CMMT plant disease dataset used in this research.

7.3.1 *Sensitivity to Hyperparameters*

Another possible reason for BYOL’s lower effectiveness in this work is its sensitivity to hyperparameter settings. Both SimCLR and BYOL can perform well with smaller batch sizes when key parameters are properly tuned. For SimCLR, stable results have been observed when adjusting temperature, learning rate, and projection head size (Bordes et al., 2023). BYOL, however, is more dependent on optimizer settings such as learning rate, weight decay, and exponential moving average (Y. Tian et al., 2021).

While techniques such as gradient accumulation and momentum encoders can help stabilize training (Grill et al., 2020), Y. Tian et al. (2021) note that BYOL is particularly sensitive to suboptimal parameter settings,

which may hinder its ability to learn meaningful representations. However, it is uncertain how much this sensitivity impacts different experimental setups, including the one used in this study. In this thesis, neither self-supervised method underwent extensive hyperparameter tuning, which likely contributed to BYOL’s lower performance.

7.3.2 Sensitivity to Data Augmentation

Both original studies (Chen et al., 2020; Grill et al., 2020) emphasize the critical role of data augmentation in their respective SSL methods. SimCLR depends heavily on strong augmentations such as random cropping, color jitter, and Gaussian blur to learn effective representations. In contrast, BYOL demonstrates greater robustness to the removal of certain augmentations. For example, excluding color distortions leads to a sharp drop in SimCLR’s performance, but it has a much smaller effect on BYOL’s predictive power.

However, the effectiveness of such augmentations is highly task-dependent. Techniques like color jitter may improve performance in some classification tasks but can hinder results in other downstream domains (Bordes et al., 2023). Similarly, Ericsson et al. (2022) demonstrate that different augmentation strategies induce the learning of distinct invariances, whose usefulness varies with the target application.

This insight could partly explain BYOL’s lower effectiveness in present research. Its original transformation pipeline, designed for ImageNet and retained here for benchmarking consistency, might not have aligned well with the characteristics of the CMMT dataset.

This discussion illustrates the need to align augmentation pipelines and training configurations with the nature of the downstream task, as characteristics unique to the target domain may limit the transferability of settings developed on standard benchmarks.

7.4 Sub-Question 3: Label Efficiency of SimCLR and BYOL Compared to Supervised Learning

The label efficiency results from this study show that supervised learning achieved the highest overall performance, with SimCLR closely following. SimCLR reached a macro F1-score of approximately 0.68 using only 5% of labeled data, while BYOL started considerably lower at 0.25 and improved gradually, reaching around 0.70 with 80% of the data labeled. In comparison, Bunyang et al. (2023) reported that both supervised and SimCLR pretrained models achieved only around 40–50% accuracy when trained with 5% and 10% of labeled data. The stronger performance of

SimCLR in this thesis at low label fractions may be due to several factors, including ImageNet pretraining, extensive hyperparameter tuning, and the use of a modern backbone architecture such as ConvNeXt-Tiny. Despite methodological differences, both studies confirm the consistent trend that model performance improves steadily as the proportion of labeled data increases.

7.5 *Scientific and Societal Impact*

This thesis demonstrates scientific relevance by being the first to compare SimCLR and BYOL for plant disease detection using a consistent architecture, dataset, and training setup. It addresses a key gap in self-supervised learning research for this domain, where standardized benchmarks are lacking. The results show that SimCLR outperforms BYOL in performance and label efficiency, providing valuable insights and a reproducible foundation for future work in plant health monitoring.

In parallel, in areas where annotated data is limited, methods like SimCLR and, to a lesser extent, BYOL can still perform well even with small training sets, as shown by the results of this thesis. When more data is available, SimCLR approaches the performance of supervised learning, making it a practical option for early disease detection. This is especially valuable in precision farming, where large volumes of unlabeled drone imagery are already being collected. Integrating these models into drone systems or automated machines could make crop health monitoring more efficient and accurate.

7.6 *Limitations and Future Direction*

While this study provides valuable insights, several limitations should be acknowledged. First, no hyperparameter tuning was performed for the self-supervised methods. Since tuning can significantly influence downstream performance and feature quality, future work should explore optimization strategies for SimCLR and BYOL in plant disease classification. The augmentation strategies used here also merit further investigation to assess whether alternative configurations could enhance results.

Second, computational constraints limited pretraining to smaller batch sizes and shorter durations. Although SimCLR and BYOL still produced reasonable results, training remained computationally expensive. Future studies could examine whether scaling up batch sizes and training time leads to improved outcomes, possibly aligning more closely with original benchmarks.

In addition, this work employed solely ConvNeXt-Tiny as the backbone. Although it performed well, evaluating other architectures such as EfficientNet or DenseNet could help determine whether patterns observed in supervised learning hold under self-supervised setups. In our benchmarks, EfficientNet performed similarly to ConvNeXt-Tiny, raising the question of how differently SimCLR and BYOL would transfer to a model with distinct architectural characteristics. Exploring this could provide deeper insight into model robustness and generalizability.

Another limitation involves consistent misclassifications within tomato and maize disease classes, as seen in confusion matrices and t-SNE plots. Although the best-performing supervised model improved maize classification, tomato diseases remained difficult to separate due to subtle visual similarities. These findings highlight a common challenge in fine-grained recognition, where convolutional models often struggle to distinguish visually similar categories (X. Zhang et al., 2022).

Future work could address this by analyzing class overlap more closely and incorporating techniques like attention mechanisms or self-reconstruction into contrastive learning. Such approaches have enhanced fine-grained feature discrimination and may improve the classification of closely related disease types (Dippel et al., 2022).

8 CONCLUSION

This thesis compares SimCLR and BYOL, two prominent self-supervised learning methods applied to plant disease classification using CNNs. Using the ConvNeXt-Tiny architecture and the CCMT dataset, the study examined contrastive and non-contrastive pretraining strategies under limited labeled data and evaluated their label efficiency. Although BYOL demonstrated performance superior to SimCLR on benchmarks such as ImageNet-1K and CIFAR-10 in previous studies, this observed the reverse trend in plant disease classification, where SimCLR consistently delivered stronger results. Even with only 5% to 10% of labeled data, SimCLR delivered solid results, while BYOL required more extensive training data to approach similar performance. Though the fully supervised ConvNeXt Tiny model achieved the highest macro F1-score, the SimCLR-pretrained model remained competitive. An ablation experiment showed that combining SimCLR with ImageNet-1k initialization could even exceed the supervised model’s test outcome, demonstrating the benefit of merging transfer learning with self-supervised training. BYOL’s comparatively reduced effectiveness in this study highlights the importance of careful hyperparameter and augmentation tuning in self-supervised learning, especially within plant disease detection. t-SNE visualizations supported

the quantitative findings, showing clear feature separation for the supervised and SimCLR models, while BYOL produced more diffuse clusters. Confusion matrices further revealed persistent difficulties in distinguishing tomato diseases across all models, pointing to the need for improved techniques that can better capture subtle visual differences.

These findings highlight the potential of self-supervised learning in plant disease detection and its relevance for practical use cases such as crop monitoring with drones. As the first study to directly assess SimCLR and BYOL in this domain, it offers clear evidence that contrastive methods like SimCLR can effectively substitute supervised models when labeled data is limited. While SimCLR outperformed BYOL in this setting, BYOL still showed promising outcomes, suggesting room for improvement through further tuning. Overall, the study demonstrates the value of self-supervised approaches for advancing visual representation learning in agriculture.

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APPENDIX A SOFTWARE DEPENDENCIES

Table 7: Python Libraries, Versions, and Sources Used in the Thesis

Package	Version	Source
PyTorch	2.5.1+cu121	Paszke et al. (2019)
PyTorch Lightning	2.5.1	Falcon et al. (2020)
Lightly	1.5.20	Susmelj et al. (2021)
Optuna	4.2.1	Akiba et al. (2019)
NumPy	2.2.5	Harris et al. (2020)
Pandas	2.2.3	McKinney (2010)
Matplotlib	3.9.2	Hunter (2007)
Seaborn	0.13.2	Waskom (2021)
Joblib	1.4.2	-
Pillow (PIL)	10.4.0	Clark (n.d.)
Scikit-learn	1.6.1	Pedregosa et al. (2012)
timm	0.9.16	Wightman (2019)

APPENDIX B HYPERPARAMETERS

Table 8: Best Hyperparameters from Optuna for Each Model

Hyperparameter	DenseNet-121	ConvNeXt-Tiny	EfficientNet-Bo
Learning Rate	5.94×10^{-5}	3.22×10^{-5}	1.57×10^{-4}
FC Layers	1	2	1
Dropout	0.597	0.580	0.363
Weight Decay	3.82×10^{-6}	8.46×10^{-5}	6.13×10^{-6}
FC Hidden Dim	1024	256	1024
Optimizer	RMSprop	AdamW	AdamW

APPENDIX C ABLATION STUDY: IMAGENET-1K INITIALIZATION ON
SELF-SUPERVISED LEARNING

An ablation experiment assessed the effect of initializing self-supervised models with weights from ImageNet. This strategy builds on earlier findings by Ciga et al. (2021), who reported that such initialization could improve convergence and boost classification performance. Gldenring and Nalpantidis (2021) also noted that preloading general visual features from ImageNet boosts training efficiency, likely because basic visual patterns are already learned, letting SSL focus on domain-specific features. While this study focused on downstream results, it did not evaluate training speed. Future research should investigate whether this initialization leads to faster convergence than training from randomly initialized weights.

Table 9: Performance of ConvNeXt-Tiny under supervised, SimCLR, and BYOL pretraining, with and without ImageNet-1K initialization.

Model	F1-score	Precision	Recall	Accuracy
ConvNeXt-Tiny (Supervised)	0.9286	0.9331	0.9253	0.9415
ConvNeXt-Tiny (SimCLR)	0.8675	0.8576	0.8818	0.8786
ConvNeXt-Tiny (BYOL)	0.7721	0.7657	0.7230	0.8051
ConvNeXt-Tiny (SimCLR + ImageNet-1K)	0.9311	0.9285	0.9354	0.9376
ConvNeXt-Tiny (BYOL + ImageNet-1K)	0.8042	0.7907	0.8351	0.8217

Using ImageNet pretraining significantly enhanced the performance of both SimCLR and BYOL. SimCLR experienced the most substantial improvement, with its F1-score increasing from 0.8675 to 0.9311, representing a relative gain of approximately 7.33%. Precision, recall, and accuracy also showed consistent increases. BYOL benefited as well, though to a lesser degree, with its F1-score rising from 0.7721 to 0.8042. Despite the more modest gain, this still points to a positive influence of ImageNet initialization on non-contrastive self-supervised learning.

Remarkably, SimCLR pretrained with ImageNet weights slightly outperformed the fully supervised ConvNeXt Tiny model, achieving an F1-score of 0.9311 compared to 0.9286. This outcome suggests combining self-supervised learning with ImageNet pretraining improves performance in specialized domains such as plant disease detection. The model likely first learned generic visual patterns from ImageNet, which then supported recognizing fine-grained features in the plant disease dataset.

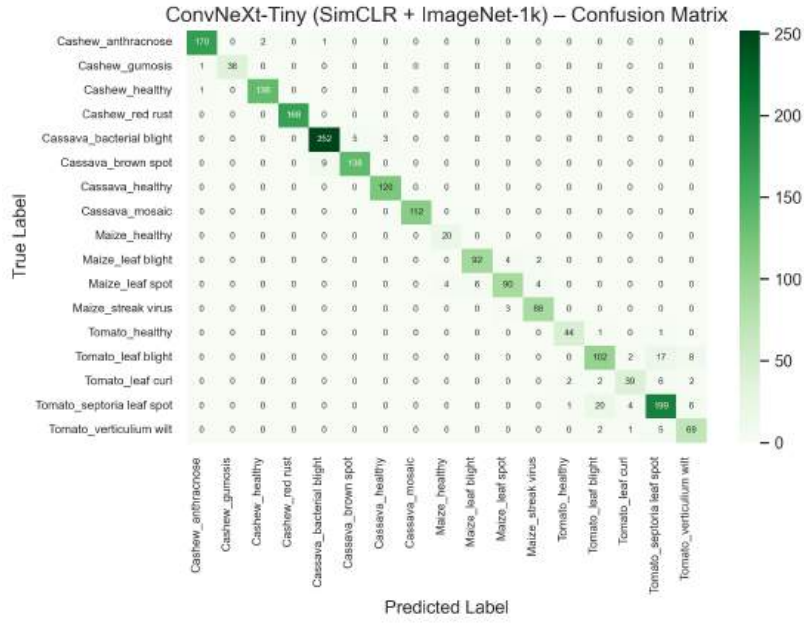


Figure 11: Confusion matrix for ConvNeXt-Tiny (SimCLR+Imagenet-1k)

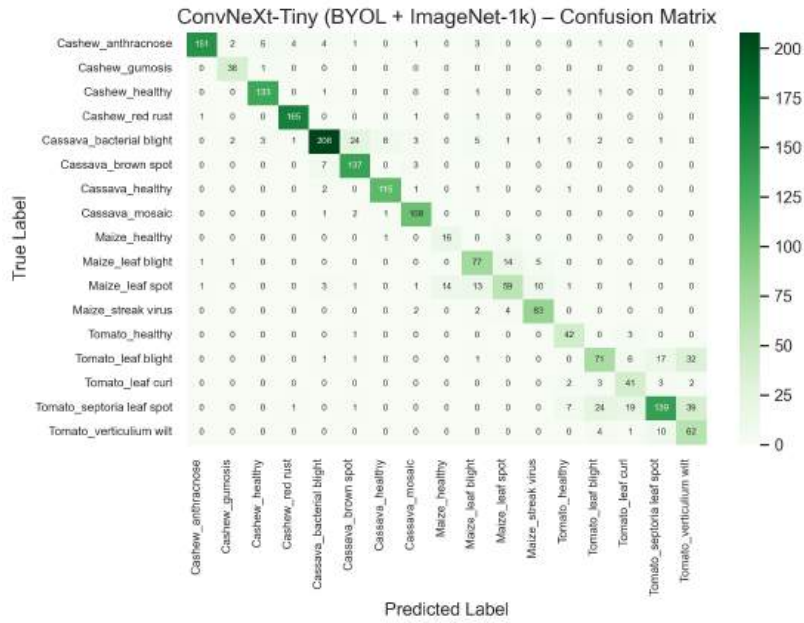


Figure 12: Confusion matrix for ConvNeXt-Tiny (BYOL+Imagenet-1k)